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Scheduled Tasks in Zoho Creator

1. Aim

To automate the repeated tasks of the RentEase application with three schedules, a daily schedule for overdue rent, a weekly schedule for the maintenance digest and a monthly schedule for rent invoices, so that these tasks run on time without anyone starting them.

2. Introduction

2.1 Schedules in Zoho Creator

A schedule runs a Deluge script automatically at a set date and time. It can be set to run once or to repeat every day, week, month or year. A schedule can be attached to a form, so that the script works on the records that match a criteria, or it can run without a form. Schedules are used for work that depends on time and not on a user action, such as reminders, reports, clean-up and recurring records.

2.2 Schedules in RentEase

Rent and repairs in RentEase depend on dates. Invoices fall due and become overdue, open repair requests must be followed up every week, and a new invoice is needed for every active lease each month. Doing these by hand is slow and easy to forget, so three schedules were created, one for each frequency.

3. Objectives of the Study

- To list the requirements that need a schedule in the rental business.
- To create a daily, a weekly and a monthly schedule in Zoho Creator.
- To use Deluge in a schedule to update records, create records and send emails.
- To verify the effect of each schedule on the application data.

4. Procedure

1. Open the RentEase application in Edit mode and click Workflow in the top bar, then open Schedules.
2. Create a new schedule for the daily overdue rent. Enter the name, select the form Rent_Invoice, set the frequency to every day and set the time.
3. In the script, fetch the invoices whose Status is Unpaid and whose Due Date has passed, set their Status to Overdue, add the late fee of Rs 500 and send an email to the tenant.
4. Create the weekly schedule for the maintenance digest. Set the frequency to every week, the day to Monday and the time to 8:00 AM. In the script, collect the open maintenance requests and email them to the manager.
5. Create the monthly schedule for the invoice generator. Set the frequency to every month, the date to the 1st and the time to 6:00 AM. In the script, create a rent invoice for each lease whose Status is Active.
6. Save the three schedules.
7. Open the application in Access mode and check the invoices, the emails and the new records that the schedules produce.

5. Detailed Analysis of the Schedules

Table 1 lists the requirements of the rental business and the schedule that meets each one.

No.	Requirement	Schedule	Frequency
1	Unpaid invoices that pass the due date must be marked Overdue, charged a late fee of Rs 500 and the tenant must be told	Overdue rent	Daily
2	The manager must receive a list of the open repair requests at the start of every week	Maintenance digest	Weekly, Monday 8:00 AM
3	Every active lease must get a rent invoice at the start of each month	Invoice generator	Monthly, 1st at 6:00 AM

Table 1: Schedule requirements of RentEase

Table 2 compares the settings of the three schedules.

Setting	Daily overdue rent	Weekly maintenance digest	Monthly invoice generator
Frequency	Every day	Every week	Every month
Run time	Time set in the schedule	Monday, 8:00 AM	1st of the month, 6:00 AM
Form	Rent_Invoice	Maintenance_Request	Lease_Agreement
Records used	Unpaid invoices past the due date	Open requests	Leases with Status Active
Action	Marks Overdue, adds Rs 500 late fee	Emails the digest	Creates an invoice
Email sent to	Tenant	Manager	None

Table 2: Settings of the three schedules

Table 3 describes the logic of each schedule in steps.

Schedule	Logic
Daily overdue rent	Step 1: fetch the Rent_Invoice records with Status Unpaid and a Due Date before today. Step 2: for each record, set Status to Overdue and Late Fee to 500. Step 3: send an email to the tenant.
Weekly maintenance digest	Step 1: fetch the Maintenance_Request records that are not closed. Step 2: build a list of the requests with their status. Step 3: send the list to the manager by email.
Monthly invoice generator	Step 1: fetch the Lease_Agreement records with Status Active. Step 2: for each lease, create a Rent_Invoice record for the new billing month with the rent of the lease.

Table 3: Logic of the schedules

Because only invoices with Status Unpaid are selected by the daily schedule, an invoice that has become Overdue is not selected again, so the late fee of Rs 500 is added once. The Total Amount of the invoice is the Rent Amount plus the Late Fee, so it rises to include the fee automatically.

6. Output

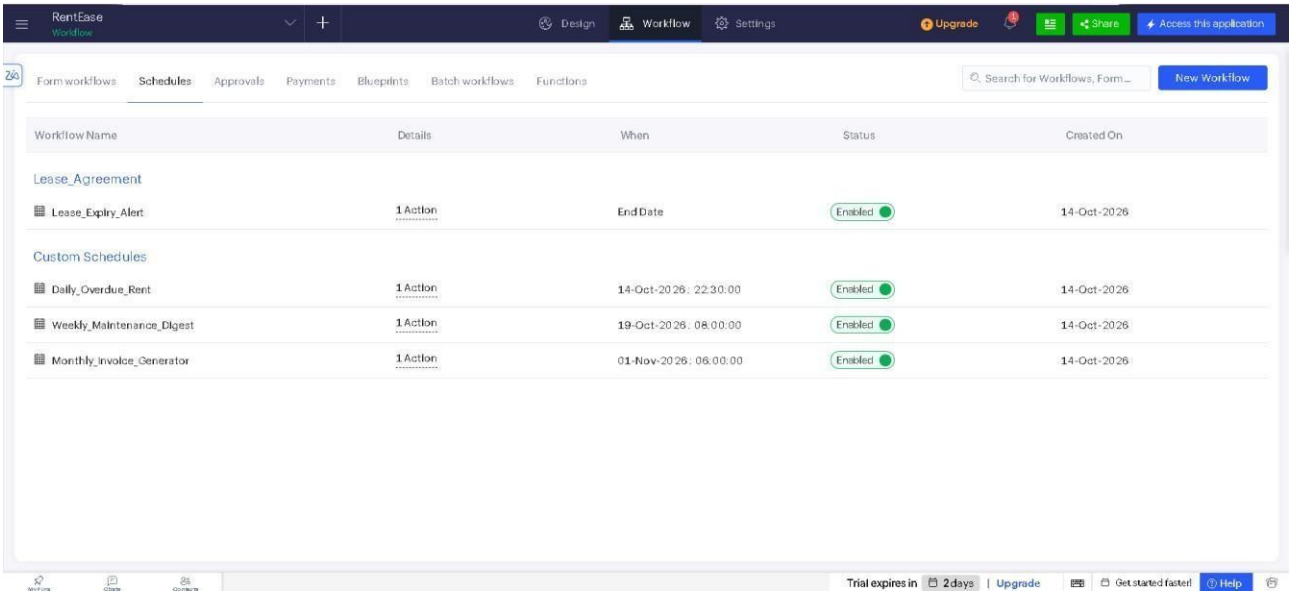


Figure 1: Schedules of RentEase



Figure 2: Daily overdue rent schedule

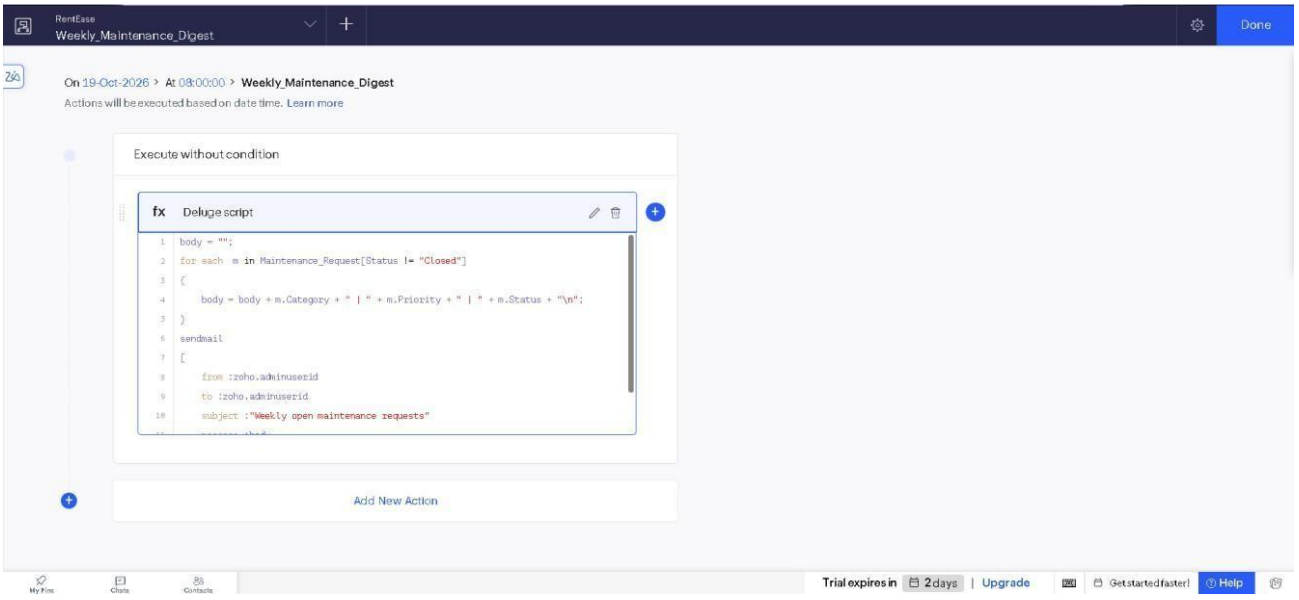


Figure 3: Weekly maintenance digest schedule

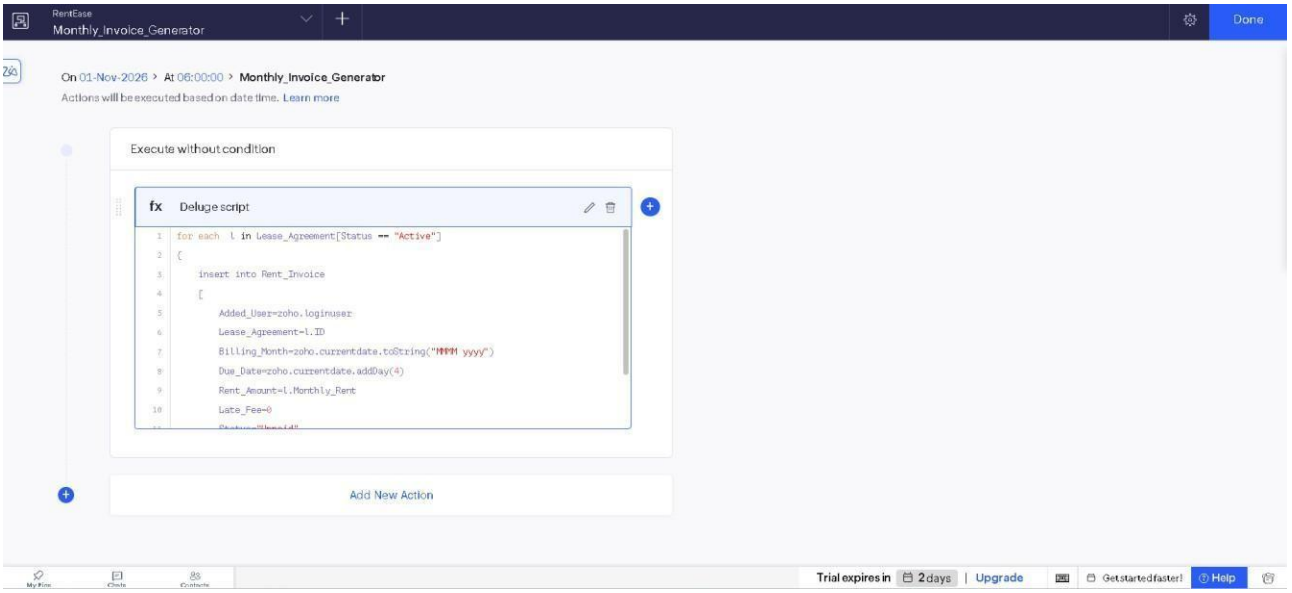


Figure 4: Monthly invoice generator schedule

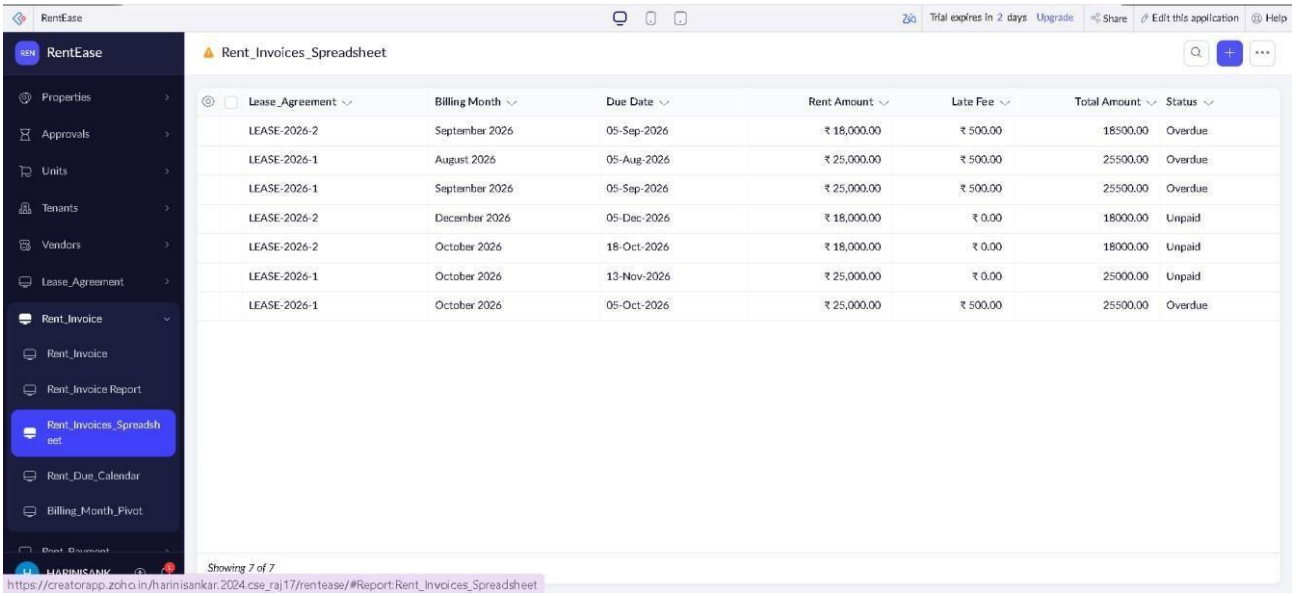


Figure 5: Overdue invoices with the late fee added by the schedule

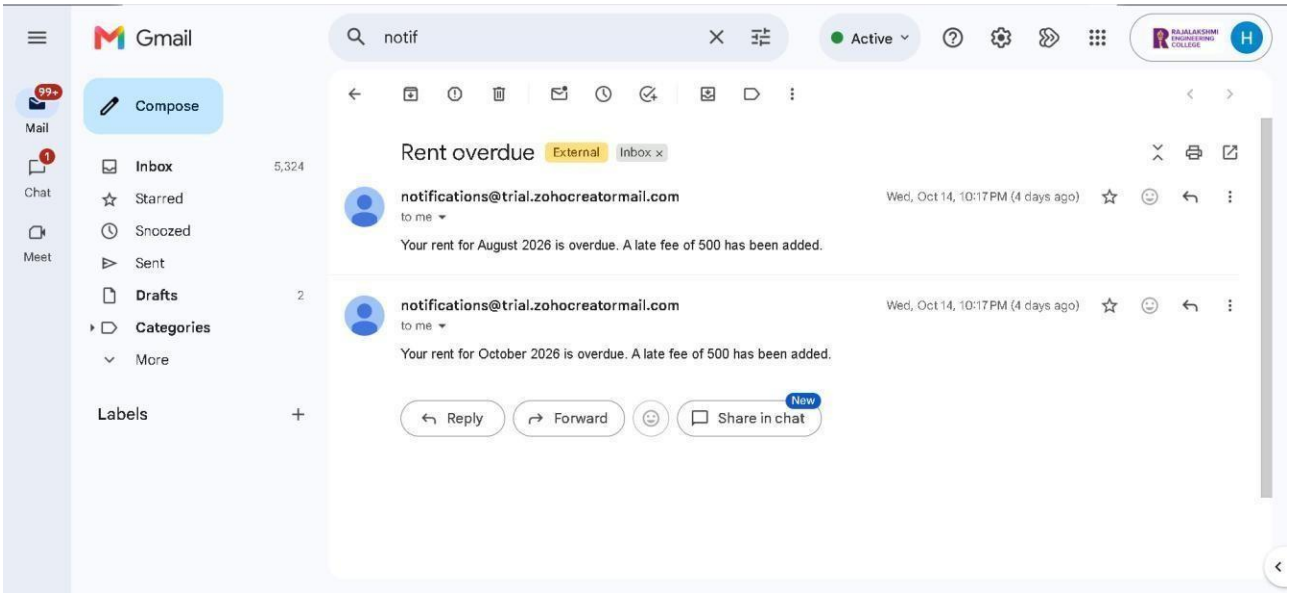


Figure 6: Email sent by a schedule

7. Result

The daily, weekly and monthly schedules of RentEase were created and checked. The daily schedule marks unpaid invoices as Overdue, adds the late fee of Rs 500 and notifies the tenant, the weekly schedule emails the open maintenance requests to the manager every Monday at 8:00 AM, and the monthly schedule creates an invoice for every active lease on the 1st at 6:00 AM. The repeated rental tasks now run on time without manual work.